

TRACE: A Traceable Agentic Co-Optimizer for Experimental Optimization

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Abstract

Low-budget experimental optimization requires making consequential experimental decisions from a small number of observations. Bayesian optimization (BO) and related black-box optimizers provide strong proposal mechanisms, but they often hide the intermediate decisions that matter to experimental users. Direct LLM recommendation can leverage broad domain priors and natural-language reasoning, but is difficult to control, audit, or align with the statistical search guarantees of the underlying optimizer. We introduce TRACE, a traceable agentic co-optimizer that keeps the optimizer as the search backbone while wrapping it in an auditable agentic decision layer. TRACE represents agent interventions as bounded actions backed by reusable skills, organizes external and experiential evidence through an Evidence & Skill System, and records every optimization step as a decision trace. On low-budget reaction-optimization benchmarks, TRACE improves over the corresponding BO-only optimizers and over LLM-assisted and tree-search baselines on most evaluated metrics. These gains suggest that our agentic system can effectively support experimental optimization by providing structured, traceable decision support around its actions, evidence use, and intervention logic.

1 Introduction

Low-budget experimental optimization is a central problem in scientific workflows. In chemistry, materials science, biology, and drug discovery, researchers often need to identify high-performing conditions or designs under expensive, slow, and noisy experimental feedback (Shields et al., 2021; Taylor et al., 2023; Tom et al., 2024; Abolhasani and Kumacheva, 2023; Seifrid et al., 2022; Chituri et al., 2024; Roohani et al., 2025). This setting is increasingly important in closed-loop and self-driving laboratory workflows, where each decision consumes limited resources. Unlike ordinary

recommendation problems, scientific experimental optimization involves heterogeneous variables with domain meaning, such as reaction conditions, material formulations, protein sequences, or process parameters. A useful optimizer therefore cannot merely return a high-scoring candidate; it must expose a decision process that researchers can inspect, challenge, and steer when necessary.

Black-box optimizers such as Bayesian optimization (BO), bandit methods, tree search, and finite-pool optimizers provide a strong search backbone for this setting (Jones et al., 1998; Shahriari et al., 2015; Srinivas et al., 2009; Wang et al., 2020). They can build statistical models from scarce observations, balance exploration and exploitation, and propose promising candidates under a limited budget. In practice, however, scientific optimization involves decisions that are not fully captured by a single acquisition score. Researchers must decide how to allocate the initial budget, whether an early promising route should be locally refined, whether a variable axis has been under-explored, and when external evidence should enter the loop. Standard optimizers expose the numerical search procedure, but they rarely represent these intermediate budget-allocation decisions as an explicit, traceable decision chain.

Large language models (LLMs) provide a complementary approach to this problem. They can interpret task descriptions, reason over scientific context, call tools, and produce human-readable rationales (Boiko et al., 2023; M. Bran et al., 2024; Yao et al., 2022; Schick et al., 2023). Recent work has explored LLMs for experimental design, candidate recommendation, reaction-condition optimization, and LLM-assisted Bayesian optimization (Roohani et al., 2025; Liu et al., 2024; Ramos et al., 2023; Han et al., 2025; MacKnight et al., 2025; Akke et al., 2025). However, directly using an LLM as an optimizer or as the final candidate recommender is difficult to control (Gupta et al., 2025;

Akke et al., 2025; Kristiadi et al., 2024). LLM recommendations can be sensitive to prompts, candidate representations, context length, and formatting choices; they do not naturally provide the uncertainty estimates that make Bayesian optimization effective. If an LLM is allowed to override the optimizer at every step, the workflow risks replacing a statistical black box with a language-model black box. The key question is therefore *how an LLM should be embedded into an existing optimization loop*, rather than whether it should replace the optimizer.

Existing LLM-assisted optimization systems often rely heavily on prompt engineering and remain task-specific (Liu et al., 2024; Ramos et al., 2023; Han et al., 2025; MacKnight et al., 2025; Patel et al., 2026; Zhang et al., 2025). A new dataset, optimizer, or experimental objective may require new prompts, context construction, candidate filtering rules, and failure-analysis logic. More fundamentally, many systems lack a reusable abstraction for the LLM’s role that specifies where it intervened, what evidence it used, and whether the intervention respected the optimizer’s decision boundary. Without such a structure, LLM assistance is difficult to reproduce, audit, and extend beyond a single benchmark pipeline (Raji et al., 2020).

To address these challenges, we introduce **TRACE** (**T**raceable **A**gentic **C**o-optimizer for **E**xperimental optimization), which treats low-budget optimization as a structured sequential decision process rather than as a sequence of isolated candidate recommendations. The existing optimizer remains the search backbone: it proposes candidates or shortlists from the current observations. The LLM agent acts as a bounded co-optimization layer around that backbone. At each iteration, the agent reads the current state, diagnoses the search state, selects a structured action, calls the relevant skills, retrieves scoped evidence, and records the intervention as a decision trace. The goal is not to delegate the final decision to the LLM, but to make its support for the optimization loop explicit, configurable, and auditable.

TRACE is built around four connected design choices. First, *actions* provide a fine-grained intervention language in which each action specifies where the agent may intervene, what context and skills it may access, and what must be recorded. Second, an *evidence & skill system* supplies governed domain knowledge and experiential evidence alongside reusable procedural skills. Third, an

agentic harness controls what the agent can see and do at each step, following a progressive disclosure principle. Fourth, a *decision trace* records the full decision context at every iteration, connecting the optimizer proposal, the agentic intervention, and the observed outcome.

We instantiate and evaluate TRACE in low-budget chemical reaction optimization, a representative experimental optimization domain with structured candidate spaces, limited budgets, and established baselines such as random search, BO-only optimizers, BO-ICL, and LA-MCTS (Shields et al., 2021; Taylor et al., 2023; Balandat et al., 2020; Hickman et al., 2025; Ramos et al., 2023; Wang et al., 2020). Across Arylation, Suzuki, and representative Buchwald subsets, TRACE improves early optimization efficiency and, in several settings, final best-so-far performance when paired with existing optimizers. The resulting agent traces allow us to attribute these gains to different forms of agentic support, including early local route formation, candidate supplementation, and governed evidence use.

Our contributions can be summarized as follows:

- We propose TRACE, a traceable agentic co-optimizer that reframes the LLM as a bounded decision layer for experimental optimization rather than an unconstrained LLM recommender.
- We introduce an agentic design based on actions, an evidence & skill system, a controller harness for context and authority control, and decision traces, making each LLM intervention bounded, evidence-backed, and auditable, while keeping the design extensible to new actions and evidence sources.
- We demonstrate improved performance on low-budget reaction-optimization benchmarks, and use trace-level audits to attribute the observed gains to specific forms of agentic support.

An anonymous repository containing the TRACE implementation, experiment configurations, analysis scripts, and audit-view generation tools is provided for review: <https://anonymous.4open.science/r/TRACE-83B6>.

2 Related Work

2.1 Black-Box Optimization

Bayesian optimization is a standard framework for sequential optimization under expensive evaluations, from early extremum-seeking methods to GP-UCB-style acquisition rules (Moćkus, 1974; Jones et al., 1998; Srinivas et al., 2009; Shahriari et al., 2015). It has become a widely used tool for reaction-condition optimization and closed-loop scientific experimentation, where each evaluation can require substantial experimental cost (Shields et al., 2021; Taylor et al., 2023; Hickman et al., 2025; Runqiu et al., 2025; Abolhasani and Kumatheva, 2023; Seifrid et al., 2022). Tree-search methods such as LA-MCTS provide an alternative when the search space benefits from explicit partitioning (Wang et al., 2020). These methods provide the statistical backbone for low-budget search, but they typically expose only the numerical search procedure and final candidate choices. In experimental workflows, however, users often also care about intermediate decisions: why a route is preserved, whether a local scan is justified, whether an axis has been under-explored, or how a failed step should influence later rounds. These decisions motivate an optimization layer that remains compatible with black-box optimizers while making the surrounding decision process inspectable.

2.2 LLM-Assisted Optimization

Recent work has investigated whether LLMs can serve as experimental designers. LLM-assisted optimization methods can be roughly grouped by the role the LLM plays in the optimization pipeline. Recommendation-driven methods ask the LLM to generate or rank candidates directly (Liu et al., 2024; Ramos et al., 2023; Yang et al., 2024; Agarwal et al., 2025). Representation-driven methods use LLM embeddings for downstream BO or nearest-neighbor selection (Nguyen et al., 2024; Ranković and Schwaller, 2023; Kristiadi et al., 2024). Surrogate-oriented methods use LLMs to augment prediction or representation learning (Ranković and Schwaller, 2025; Han et al., 2025; Patel et al., 2026; Zhang et al., 2025). Separately, tool-using scientific agents offload operations to scoped tools rather than relying on the LLM to infer from raw context, improving controllability and making system behavior easier to inspect (Boiko et al., 2023; M. Bran et al., 2024; Akke et al., 2025; Gupta et al., 2025; Schick et al., 2023; Yao et al.,

2022). TRACE follows this tool-mediated principle, but differs from direct recommendation and surrogate-augmentation approaches by treating the LLM as a bounded intervention layer around an existing optimizer. Each intervention is expressed as an action, constrained by scoped evidence and skills, and recorded in a per-iteration decision trace.

2.3 BO with LLMs in Chemistry

Reaction optimization provides a concrete evaluation setting for LLM-assisted BO, as reaction spaces are structured, knowledge-intensive, and costly to evaluate. LLM-GO evaluates fully enumerated categorical reaction datasets and finds that LLM-guided optimization can match or exceed BO when high-performing conditions are scarce, while BO remains preferable for continuous or multi-objective settings (MacKnight et al., 2025). ChemBOMAS also combines LLMs, chemical evidence, and BO for chemistry (Han et al., 2025). Its mechanism centers on a training-augmented LLM regressor, pseudo-data, and a knowledge-guided hierarchical search tree. In contrast, TRACE retains the BO optimizer as the search backbone and wraps it with agent-side interventions whose scope, evidence, and rationale are recorded in a per-iteration decision trace. This design targets a different role for language models: not replacing or retraining the optimizer, but making bounded support around the optimizer explicit, configurable, and auditable.

3 Method

3.1 Formulation

We frame low-budget experimental optimization as a structured sequential decision process rather than a single-step candidate-recommendation problem. A method must choose a sequence of experiments from a structured candidate pool under a small budget, and TRACE makes the intermediate budget-allocation decisions in this process explicit and traceable. Let $\mathcal{X}$ denote a finite candidate pool. Before iteration t , the observed data are

$$\mathcal{D}_{t-1} = \{(x_i, y_i)\}_{i=1}^{t-1}, \quad x_i \in \mathcal{X}, \quad y_i = f(x_i), \quad (1)$$

where f is the experimental response or, in an offline benchmark, the finite-pool oracle. The optimization objective is to improve the best-so-far value

$$b_t = \max_{i \leq t} y_i. \quad (2)$$

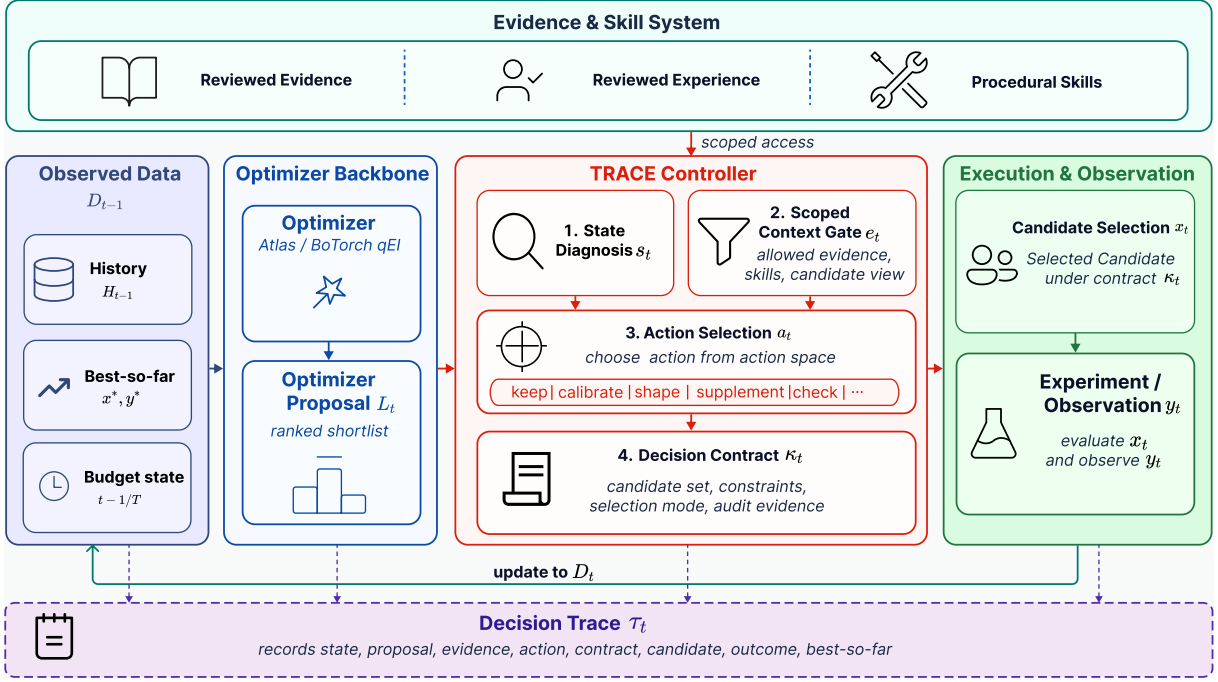


Figure 1: Overall architecture of TRACE. The optimizer remains the search backbone and proposes ranked candidates from the observed data, while the TRACE controller acts as a bounded agentic decision layer rather than a standalone LLM recommender. The controller activates scoped evidence and allowed skills, selects a structured action, and converts the optimizer proposal into a bounded decision contract for final candidate selection. Each step is recorded as a decision trace containing the action rationale, evidence use, contract, candidate, and outcome for inspection and audit.

The numerical optimizer remains the search backbone. At each iteration, a BO optimizer P , instantiated by Atlas or a BoTorch qEI implementation, proposes a ranked candidate list or shortlist (Hickman et al., 2025; Balandat et al., 2020),

$$L_t = P(\mathcal{D}_{t-1}, \mathcal{X} \setminus \{x_i\}_{i < t}). \quad (3)$$

TRACE uses this optimizer proposal as the starting point for each decision. The TRACE controller mediates between the ranked proposal and the final experimental choice. It diagnoses the current optimization state, selects a structured action for the current decision, converts the action into a bounded decision contract, and records the resulting decision. This makes the optimizer part of a traceable decision process rather than a hidden loop that only exposes the final selected candidate.

3.2 TRACE Controller

The TRACE controller is the runtime decision layer around the optimizer backbone. At iteration t , the controller selects an action conditioned on the current state s_t , the optimizer proposal L_t , and the scoped evidence e_t available at that iteration:

$$a_t = C_\theta(s_t, L_t, e_t). \quad (4)$$

We use s_t to denote a structured summary of the optimization state at iteration t , including recent observations, the current best-so-far b_{t-1} , and simple stagnation indicators. The parameter θ collects the fixed components that define the controller, including the prompt template, action schema, optimizer-aware configuration, guardrails, and the set of available skills.

The controller also serves as the harness for each decision point. It manages state summaries, optimizer proposals, candidate details, evidence access, skill availability, and authority boundaries. Its role is not to expose more information to the LLM, but to restrict each decision point to the context that the current action and phase require. Following a progressive disclosure principle (Sumers et al., 2023; Akke et al., 2025), full histories, candidate pools, knowledge cards, and skill specifications are not placed in the context by default. Instead, the controller reveals information according to the current action, optimization phase, and optimizer-aware configuration. In practice, this reduces token cost over multi-step runs, limits distraction from irrelevant or stale evidence, and makes agent behavior easier to configure and audit.

Operationally, each controller step follows the same bounded execution path. The LLM selects from a fixed action space, and the action output must be parsed into the contract form used by the downstream selector. Outputs that do not match the action schema, request unavailable information, or exceed the configured authority boundary are rejected and fall back to the optimizer-only selection path. The accepted action, contract, fallback status, and selected candidate are then written to the decision trace.

3.3 Agentic Actions

Actions provide the fine-grained intervention language by which the agent operates within the optimization loop. Without actions, an LLM tends to produce free-form advice or direct candidate recommendations, making it unclear whether the suggestion is intended to change initialization, reshape a shortlist, supplement candidates, review risk, or merely explain the current state. TRACE instead requires the agent to choose from structured actions with explicit intervention points, allowed context, callable skills, authority boundaries, and audit requirements.

The output of an action is not an unconstrained candidate. It is a bounded decision contract,

$$\begin{aligned}\kappa_t &= \phi_{a_t}(L_t, \mathcal{X}, \mathcal{D}_{t-1}, e_t), \\ \kappa_t &= (\mathcal{C}_t, r_t, m_t, \xi_t).\end{aligned}\quad (5)$$

The contract specifies the candidate set or shortlist $\mathcal{C}_t$, risk and authority constraints r_t , a selection mode m_t , and audit evidence ξ_t . For example, an action may keep the optimizer’s top candidate, perform a local calibration around an early promising route, shape a shortlist for better coverage, or supplement the optimizer shortlist with a small number of evidence-supported candidates. The final experiment is then selected from the contract:

$$x_t = \text{Select}(\kappa_t). \quad (6)$$

Actions thus convert LLM judgment into structured, bounded, and auditable interventions in the optimization loop. The LLM does not freely override the optimizer; it changes how optimizer proposals, candidate sets, and evidence are used within an explicit decision boundary. The full per-iteration action decision flow is illustrated in Figure A1 (Appendix).

3.4 Evidence & Skill System

Scientific experimental optimization often depends on external evidence: papers, dataset descriptions, domain background, and reviewed experience from prior analyses. Exposing such information to an LLM is useful but risky. Raw text can pollute the context, evidence may apply only to a narrow setting, and benchmark-derived experience can leak task-specific information into a formal comparison. TRACE organizes external support as an Evidence & Skill System rather than as an unstructured retrieval pipeline (Lewis et al., 2020; Gao et al., 2023; M. Bran et al., 2024).

The evidence system stores reviewed evidence with source, scope, activation stage, allowed usage, and leakage-risk metadata. It includes reviewed domain knowledge from literature, web sources, dataset descriptions, or scientific background. When trajectory-derived experience is used, it is stored as scoped, non-oracle guidance rather than as candidate-level outcome lookup or ranking information. We treat such reviewed experience as a separate governance tier with stricter source and scope checks than reviewed domain knowledge, so that it can enter runtime decisions only under matching scope constraints and allowed-use policies. The skill system provides procedural capabilities used by actions: initial design, stagnation diagnosis, candidate probing, shortlist reranking, local calibration, risk checking, literature search, PDF extraction, evidence filtering, knowledge-card construction, and trace attribution. Evidence explains why an intervention is supported; skills define how it is executed or governed.

At runtime, the controller activates only scoped evidence for the current decision:

$$e_t = R(q_t, \mathcal{E}; \Omega_t). \quad (7)$$

Here $\mathcal{E}$ is the reviewed evidence store, q_t is a question induced by the current state or action need, and Ω_t defines the applicable scope, including dataset, optimizer, optimization phase, allowed action types, and allowed-use constraints. This prevents the evidence system from becoming a global prompt prefix. Evidence enters the controller only when it is relevant to the current decision and allowed by the run configuration. In the offline finite-pool benchmarks, unobserved outcomes are not available to the controller or evidence retriever during optimization. The controller observes an

outcome only for the candidate selected at that iteration. Unobserved finite-pool outcomes and oracle-level summary statistics are used only after a run is complete to compute aggregate evaluation metrics.

TRACE also supports question-driven evidence acquisition. Rather than collecting all documents that mention a task, the system first identifies what evidence could support upcoming decisions, then produces reviewed cards through search, academic retrieval, and PDF extraction. These cards can later be activated through $R(q_t, \mathcal{E}; \Omega_t)$.

3.5 Decision Trace

Each iteration records the full decision context, including the optimization state, optimizer proposal, scoped evidence, selected action, decision rationale, decision contract, final candidate, observation, and best-so-far value:

$$\tau_t = (s_t, L_t, e_t, a_t, \rho_t, \kappa_t, x_t, y_t, b_t). \quad (8)$$

Here ρ_t denotes the agent’s natural-language rationale for the diagnosis, action choice, and candidate-selection decision. The full trajectory is

$$\mathcal{T}_{1:T} = \{\tau_t\}_{t=1}^T. \quad (9)$$

This trace object connects the numerical optimizer, the evidence system, the agentic action, the agent’s rationale, and the observed outcome. Because the rationale is generated as part of the decision process, the trace exposes not only what was selected, but also why the controller selected a particular action, which evidence or skills supported it, and how the final candidate was obtained from the optimizer proposal or action contract. For failed iterations, the same trace helps separate failures attributable to the optimizer shortlist, the action trigger, skill execution, evidence scope, or the agent’s own decision.

These traces also support governance by enabling measurement of action frequency, diagnosis of failure modes, and distillation of historical trajectories into reviewed experience for the Evidence & Skill System (Raji et al., 2020). Appendix A7 shows the corresponding static HTML audit view and a structured trace artifact for a concrete run.

4 Experiments

We evaluate TRACE on low-budget reaction-condition optimization as a bounded agentic decision layer around existing BO optimizers. The evaluation uses the same finite-pool protocol and observation budget across methods, focusing on whether

TRACE improves budget use without changing the underlying optimizer.

4.1 Experimental Setup

All tasks use a finite-pool optimization protocol. At each step, a method selects one unobserved candidate from the pool and receives its ground-truth yield. We use a total budget of 20 observations, with 5 initialization points and 15 sequential optimization steps. We report cumulative best-so-far values at 5, 10, and 20 observations (best@5, best@10, best@20). best@5 mainly reflects the yield of the initial design, while best@10 and best@20 measure how the sequential optimizer uses the remaining budget. Because TRACE initialization may trade off immediate yield against coverage and later diagnosability, we treat best@5 as an initialization diagnostic rather than as a standalone measure of agentic quality. We also compute the area under the best-so-far curve (AUC) as a full-trajectory efficiency metric:

$$\text{AUC} = \frac{1}{T} \sum_{t=1}^T b_t, \quad T = 20, \quad (10)$$

where b_t is the best yield observed after t observations. This average includes both initialization and sequential optimization steps.

We evaluate on Arylation (Shields et al., 2021) (3,955 candidates), Suzuki HTE (Reizman et al., 2016) (5,760), and two Buchwald–Hartwig amination subsets derived from Ahneman et al. (2018) (792 each). Arylation tests the full agentic loop in a larger pool; Suzuki HTE tests route preservation under strong baselines; Buchwald sub1 and sub2 test smaller product-level pools with different difficulty profiles. Subset definitions and oracle statistics are in Appendix A4.

We compare against four groups of baselines. Random Search measures finite-pool random exploration. BO-only runs the underlying optimizer without the TRACE controller, with two configurations: Atlas (Hickman et al., 2025) and a BoTorch qEI implementation (Balandat et al., 2020). BO-ICL (Ramos et al., 2023) is a test-time LLM-assisted optimization baseline that we adapt to the finite-pool setting using the same gpt-5.4-mini backbone as TRACE. LA-MCTS (Wang et al., 2020) is an external tree-search baseline. All main results use five random seeds and the same optimization budget: each method observes 5 initialization points and then selects 15 additional points, for a total bud-

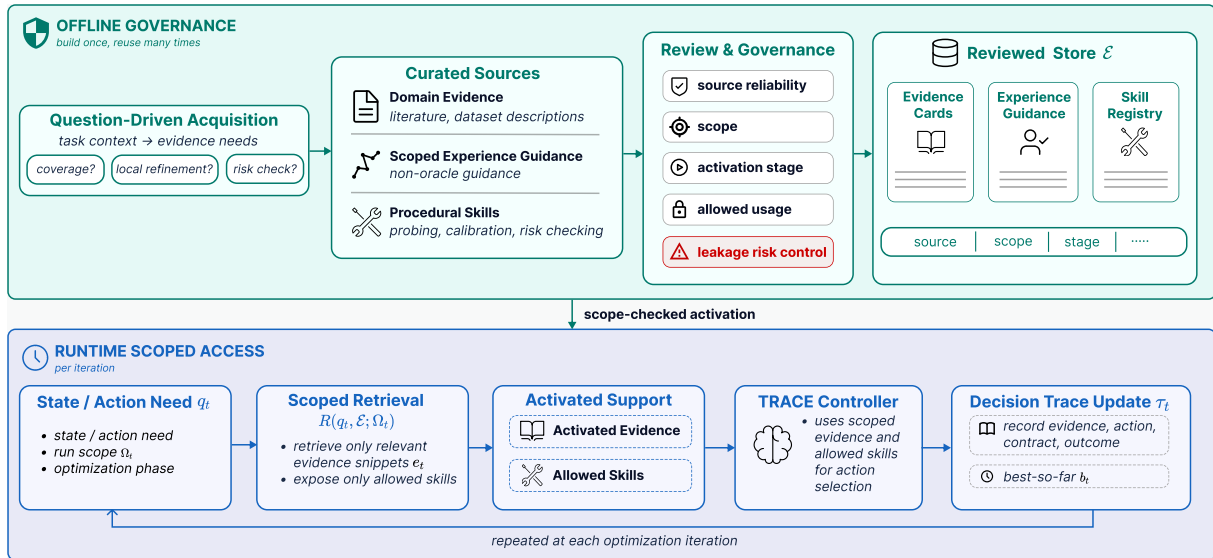


Figure 2: Question-driven Evidence & Skill System. Offline acquisition converts domain sources, dataset descriptions, reviewed experience, and procedural tools into evidence cards and skill entries with source, scope, activation stage, and allowed-use metadata. At runtime, the controller retrieves only evidence and skills relevant to the current state or action need, preventing the evidence store from becoming an unconstrained prompt prefix.

get of 20 observations. Unless otherwise specified, TRACE uses gpt-5.4-mini as its LLM backbone. We do not report ChemBOMAS (Han et al., 2025) as a direct numerical baseline because its setting depends on task-specific training, pseudo-data construction, and data-processing details that are not available for a controlled finite-pool comparison. Appendix A6 reports an LLM-backbone sensitivity analysis.

4.2 Main Results

Table 1 shows that TRACE improves low-budget optimization in several settings, with the clearest gains at best@10 and best@20. Arylation provides the cleanest full-loop comparison: TRACE improves over both BO-only optimizers and over the BO-ICL and LA-MCTS baselines. On Suzuki HTE, TRACE + Atlas achieves the highest best@10 and the second-best best@20, while TRACE coupled with the BoTorch qEI optimizer improves over the BotorchQEI-only baseline and yields the strongest AUC overall (full per-metric numbers in the appendix). The Atlas-only baseline has high variance at best@10 because a small number of seeds enter early local stagnation before recovering later; TRACE reduces this early-budget instability through route preservation and local calibration.

The Buchwald subsets show a complementary pattern. On Buchwald sub1, where the pool is closer to saturation, TRACE improves the key bud-

get checkpoints for both Atlas and the BoTorch qEI optimizer. For Buchwald sub2, whose oracle maximum is much lower than the other pools, TRACE improves both BO optimizers and gives the strongest best@20 when paired with the BoTorch qEI optimizer. Taken together, these results indicate that TRACE can improve budget use without replacing the optimizer. Its gains come from constraining, calibrating, or supplementing the optimizer’s candidate proposals within a traceable decision boundary.

4.3 Component Ablations

We study components on a clean Suzuki + Atlas ablation setting with five seeds. We select this setting because TRACE shows clear improvement while the baselines remain competitive, making component-level differences informative. The dataset, optimizer, budget, and seeds match the Suzuki + Atlas setting in Table 1; because the ablation uses a separate clean configuration, its rows are intended for within-table component comparison.

Table 2 indicates that the full TRACE configuration improves early efficiency and AUC over Atlas only. Removing reviewed knowledge reduces best@10, best@20, and AUC, suggesting that reviewed evidence improves stability in this setting. Removing LLM initialization returns best@5 to the random-initialization level used by the BO-only baselines and substantially lowers best@10 and

Method	Ary@10	Ary@20	Suz@10	Suz@20	Bsub1@10	Bsub1@20	Bsub2@10	Bsub2@20
Random	76.84 $\pm$ 9.73	86.40 $\pm$ 9.03	90.32 $\pm$ 5.97	93.06 $\pm$ 6.08	83.03 $\pm$ 12.33	89.13 $\pm$ 5.33	55.39 $\pm$ 6.48	55.39 $\pm$ 6.48
BO-ICL	83.65 $\pm$ 2.42	87.93 $\pm$ 5.82	88.93 $\pm$ 5.10	96.47 $\pm$ 2.63	88.78 $\pm$ 5.63	93.46 $\pm$ 4.46	52.11 $\pm$ 6.48	55.77 $\pm$ 4.53
LA-MCTS	78.37 $\pm$ 7.12	80.51 $\pm$ 9.37	90.71 $\pm$ 8.66	91.09 $\pm$ 8.88	90.62 $\pm$ 8.38	96.12 $\pm$ 2.54	49.97 $\pm$ 7.33	58.25 $\pm$ 5.83
Atlas only	78.98 $\pm$ 7.75	90.44 $\pm$ 6.70	75.86 $\pm$ 29.15	93.31 $\pm$ 7.05	88.85 $\pm$ 9.89	93.93 $\pm$ 5.69	51.68 $\pm$ 6.00	55.14 $\pm$ 6.02
TRACE + Atlas	<u>85.33</u> $\pm$ 14.62	94.37 $\pm$ 5.00	92.83 $\pm$ 8.00	<u>96.37</u> $\pm$ 2.99	<u>90.85</u> $\pm$ 8.38	97.23 $\pm$ 2.53	57.31 $\pm$ 9.48	<u>59.22</u> $\pm$ 8.11
BotorchQEI only	81.11 $\pm$ 8.74	88.91 $\pm$ 7.40	86.05 $\pm$ 17.15	91.93 $\pm$ 11.67	87.49 $\pm$ 14.48	95.63 $\pm$ 5.75	51.72 $\pm$ 10.14	58.03 $\pm$ 10.08
TRACE + BotorchQEI	87.36 $\pm$ 7.93	<u>93.67</u> $\pm$ 4.13	<u>90.88</u> $\pm$ 4.81	93.88 $\pm$ 2.68	95.96 $\pm$ 2.82	<u>96.78</u> $\pm$ 2.67	<u>55.56</u> $\pm$ 8.11	63.03 $\pm$ 7.00

Table 1: Main results on low-budget reaction optimization tasks (mean $\pm$ std over five seeds; total budget 20, initialization budget 5). **Bold**: best per column; underline: second best. best@5, AUC, best-so-far curves, and per-seed details are in the appendix.

Method	@5	@10	@20	AUC
Atlas only	70.39 $\pm$ 24.83	75.86 $\pm$ 29.15	93.31 $\pm$ 7.05	78.85 $\pm$ 21.72
Full TRACE	85.54 $\pm$ 15.30	93.57 $\pm$ 5.34	<u>95.67</u> $\pm$ 0.88	89.94 $\pm$ 5.13
w/o reviewed knowledge	84.09 $\pm$ 7.61	87.43 $\pm$ 4.88	94.38 $\pm$ 2.84	85.17 $\pm$ 4.28
w/o action	<u>84.41</u> $\pm$ 3.71	<u>88.59</u> $\pm$ 7.74	95.79 $\pm$ 0.79	<u>87.99</u> $\pm$ 4.91
w/o LLM init	70.39 $\pm$ 24.83	84.70 $\pm$ 18.76	93.67 $\pm$ 9.50	82.06 $\pm$ 17.47

Table 2: Component ablation on Suzuki + Atlas. Starting from the full TRACE configuration, each variant removes one major component: reviewed knowledge, action-based intervention, or LLM initialization. best@5 is included here to isolate initialization effects.

AUC, indicating that early route formation is an important source of improvement for Suzuki + Atlas. The w/o action variant remains close to the full system in aggregate metrics: AUC drops, while best@20 stays competitive. This pattern suggests that actions primarily affect localized trajectory decisions rather than uniformly shifting every aggregate metric. Their role is therefore better examined through trajectory-level analyses, where the trace exposes how online interventions reshape candidate selection.

To make this localized effect explicit, Appendix A3 provides a fixed-initialization action analysis. In that paired Buchwald sub2 run, TRACE and BO-only share the same first five observations, so the later trajectory gap isolates the effect of online action intervention.

4.4 Traceability and Audit

Each agentic run records the full per-iteration decision context described in §3.5. The record stores the optimizer proposal, selected action, decision contract, evidence and skill access, final candidate, observation, best-so-far value, and post-

observation reflection. These fields connect the numerical gains in Table 1 and Table 2 to inspectable decision records, making it possible to attribute each performance change to initialization, evidence, actions, or the optimizer.

We also render the same run artifacts into a static audit view for manual inspection. Appendix A7 shows a Buchwald sub2 run with the BoTorch qEI optimizer in which TRACE applies a coverage-oriented shortlist shaping action after initialization. The audit record shows the action timeline and expands iteration 8, where the selected candidate differs from the optimizer top-1 candidate and improves the best-so-far value from 57.09 to 66.02.

5 Conclusion

TRACE presents a traceable agentic co-optimizer for low-budget experimental optimization. Rather than replacing an existing optimizer with an LLM, TRACE decomposes optimization into auditable decisions around the optimizer, including initialization, diagnosis, bounded action selection, evidence use, and trace recording. This design keeps the numerical optimizer as the search backbone while giving the agent a controlled way to support decisions that are difficult to express as a single acquisition function. Across reaction-optimization benchmarks, TRACE improves over corresponding BO-only optimizers and over LLM-assisted and tree-search baselines on most evaluated metrics. The ablations and case study further suggest that no single component is universally responsible. Instead, different components become useful under different failure modes, and the decision trace makes these effects individually inspectable.

Limitations

This work evaluates TRACE on finite-pool reaction-optimization benchmarks, where candidate sets and oracle yields are fixed. This protocol

allows controlled comparison across optimizers, but it does not capture all constraints of a deployed wet-lab loop, such as measurement noise, failed experiments, batching, turnaround time, and heterogeneous experimental costs. Applying TRACE in such settings would require additional validation under the corresponding laboratory protocol.

The main comparisons use five seeds, and low-budget finite-pool optimization exhibits substantial seed-to-seed variance. We therefore emphasize aggregate trends together with trajectory-level analyses rather than treating any single seed or metric as definitive. We include LLM-backbone diagnostics, but do not exhaustively sweep model families, model sizes, or prompting policies.

Finally, TRACE relies on reviewed evidence, scoped action spaces, and allowed-use policies. These components make the agentic decisions more traceable, but they also require domain-specific adaptation when moving beyond the reaction-optimization tasks studied here. In new domains, the evidence sources, procedural skills, action space, and candidate representations would need to be reviewed and configured for the target experimental setting.

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A1 Per-Iteration Action Decision Flow

Figure A1 expands the per-iteration controller step in TRACE. It highlights the runtime order from state construction and action selection to contract validation, candidate selection, observation, and trace update. The LLM does not directly emit an experiment; its output is converted into a bounded contract and checked before the final candidate is selected.

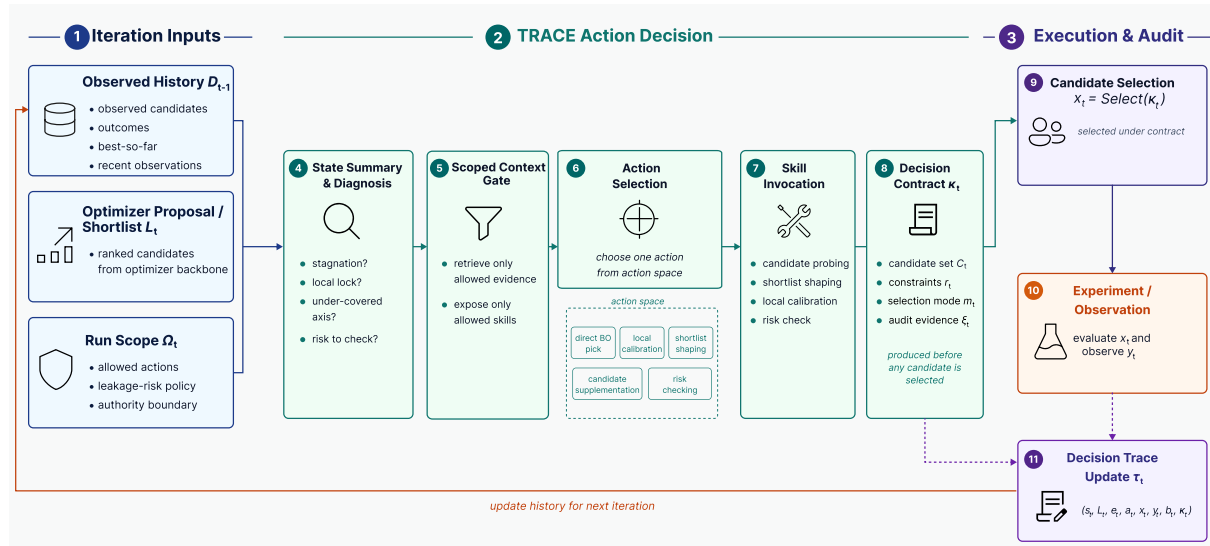


Figure A1: Per-iteration action decision flow. TRACE converts the agent’s decision into a bounded intervention over the optimizer workflow rather than unrestricted free-form recommendation.

A2 Best-So-Far Curves

Figure A2 shows the best-so-far trajectories used to compute AUC. The curves are intended as trajectory-level support for the aggregate metrics rather than as separate claims. They are most useful for comparing when a method reaches a high-yield region, especially in cases where best@20 values are close but AUC differs.

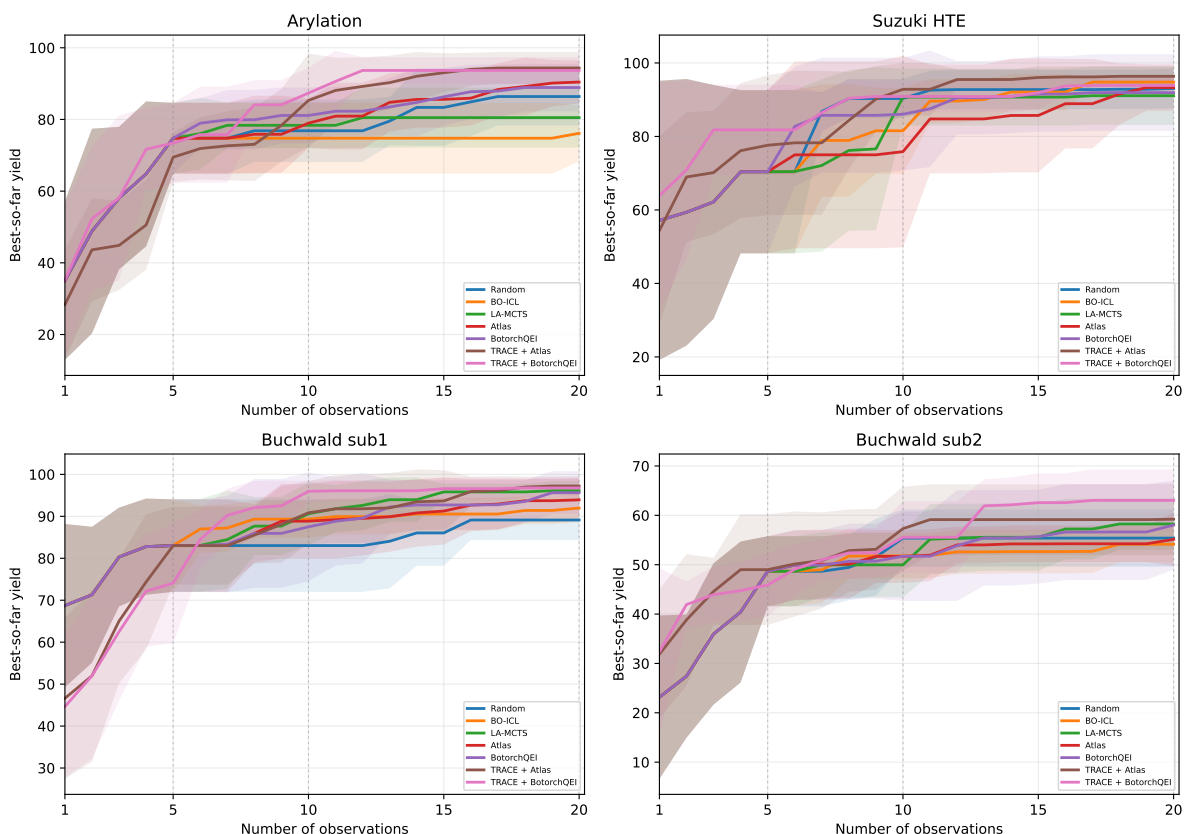


Figure A2: Best-so-far curves for the four evaluated finite-pool tasks.

Dataset	Pool size	Max	95th pct.
Arylation	3,955	100.00	83.21
Suzuki HTE	5,760	100.00	90.99
Buchwald sub1	792	100.00	91.02
Buchwald sub2	792	68.25	52.44

Table A1: Finite-pool statistics for the evaluated datasets.

A3 Fixed-Initialization Action Analysis

To isolate online action intervention from initialization, we run a paired Buchwald sub2 analysis with the BoTorch qEI optimizer. The BO-only trajectory is restarted from the same first five observations selected by TRACE, so the two trajectories are identical through best@5 and diverge only after sequential optimization begins.

In this paired run, direct BO plateaus after the shared initialization, while TRACE applies coverage-oriented shortlist shaping in several rounds. The action does not replace BO with a free-form LLM recommendation; it constrains the candidate set under an auditable policy before the optimizer selects the final candidate.

A4 Experimental Details

All experiments use the same finite-pool protocol as the main text: each method observes five initialization points and then selects fifteen additional points sequentially from the unobserved candidate pool. The reported budget is therefore 20 total observations. The main experiments use five random seeds. For methods whose initialization is not selected by an LLM (Random, BO-ICL, LA-MCTS, and BO-only), the same random initialization set is shared within each dataset and seed.

The two Buchwald subsets are full product-level pools filtered by target product. buchwald_sub1 corresponds to the product Cc1ccc(Nc2ccccc2)cc1, and buchwald_sub2 corresponds to C0c1ccc(Nc2ccc(C)cc2)cc1; each subset contains 792 candidates.

Table A1 summarizes the finite-pool statistics used to interpret absolute performance. These oracle statistics are used only for reporting and are not exposed to the optimizer or agent.

The evaluated methods are Random Search, BO-ICL, LA-MCTS, BO-only with Atlas or a BoTorch qEI implementation, and TRACE paired with the same two BO optimizers. For BO-ICL, we use gpt-5.4-mini with UCB acquisition and

text-embedding-3-small to map inverse candidate descriptions back to the finite pool.

A5 Full Quantitative Results

Table A2 reports the full metric set behind Table 1. The main paper emphasizes best@10 and best@20, while this table additionally includes best@5 and AUC.

A6 LLM-Backbone Sensitivity Analysis

Table A3 evaluates TRACE on Suzuki HTE + Atlas with three language-model backbones while keeping the optimizer, budget, prompts, and controller workflow fixed. The results are broadly stable across backbones: all three variants reach best@20 above 96, and their AUC values remain in a similar range. This suggests that the TRACE workflow is not highly sensitive to the choice of LLM backbone under this setting.

Backbone	best@5	best@10	best@20	AUC
gpt-4o-mini	75.41±34.25	92.55±6.56	96.32±4.11	82.21±9.62
gpt-5.4-mini	76.90±24.37	96.19±3.69	97.37±3.46	86.72±9.16
gpt-5.4	77.60±24.85	93.23±1.32	98.45±2.69	88.08±6.94

Table A3: LLM-backbone sensitivity of TRACE on Suzuki HTE + Atlas over three seeds with the same optimizer, controller workflow, and optimization budget as the main experiments.

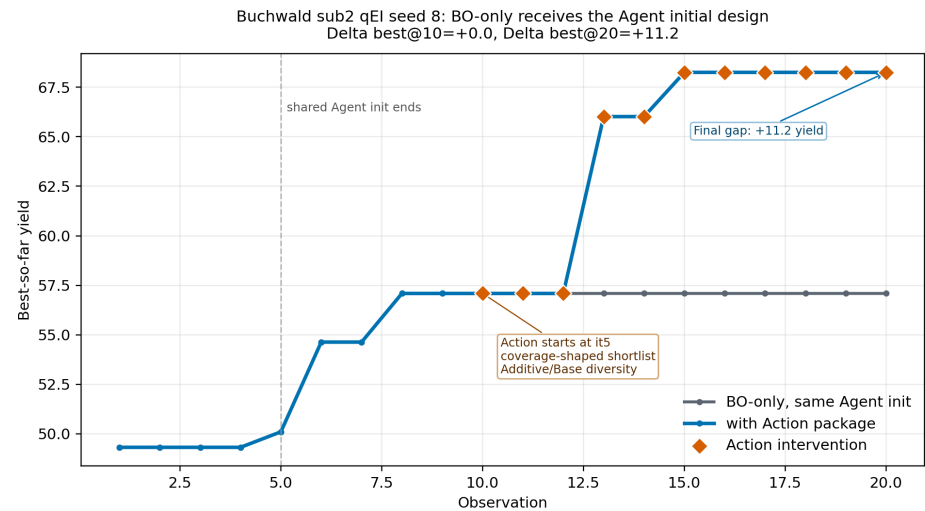
A7 Trace and Audit Example

The full audit artifact is a static HTML view generated from the run output directory, which contains the per-iteration trace, the progress log, the run summary, and the LLM usage record. It instantiates the per-iteration record τ_t defined in §3.5 for every iteration of the run.

Table A4 shows the structured counterpart to the expanded view in Figure A5. Each field corresponds to a component of the per-iteration record τ_t defined in §3.5: the optimizer proposes a qEI shortlist, the agent diagnoses uneven coverage, selects shape_only_bo_pick to rebalance the candidate set, and records the action rationale, selected candidate, evidence use, and observed outcome. The same fields appear at every iteration, making it possible to trace when and why the agent intervened.

Table A5 gives a compact multi-iteration view of the same audit case. After the first post-initialization BO picks, TRACE applies coverage-oriented shortlist shaping in iterations 5–8. The table records whether the action used evidence, whether the candidate probe was active, whether

A. Buchwald sub2 + BotorchQEI, seed 8



B. Buchwald sub2 + BotorchQEI, seed 9

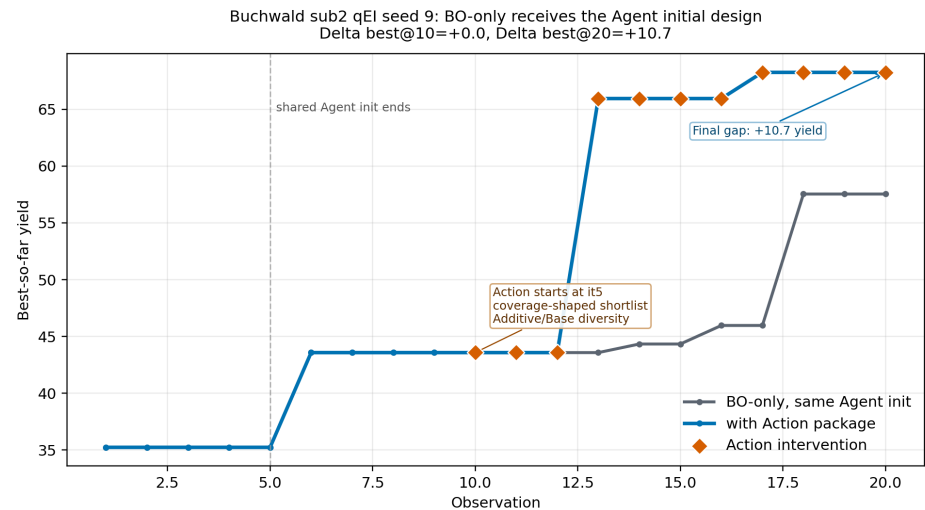


Figure A3: Action-mediated trajectory divergence under fixed TRACE initialization. The paired runs share the same first five observations; the later gap reflects online action intervention rather than initialization advantage.

the final choice differed from the optimizer top-1 candidate, and how the best-so-far value changed.

A8 Evidence and Skill Examples

Table A6 shows representative entries from the Evidence & Skill System used by TRACE: reviewed domain knowledge, reviewed experience, and procedural skills, each annotated with scope, intended stage, operational role, and a safety note. The table is intentionally abbreviated: it shows the governance fields needed to understand scope and allowed use, rather than the full internal card text.

A9 Ablation Details

The ablation in Table 2 removes reviewed evidence, action-based intervention, or LLM initialization from the Suzuki + Atlas TRACE configuration. These variants isolate the major system components rather than enumerate all internal action settings.

Dataset	Method	n	best@5		best@10		best@20		AUC	
Arylation	Random	5	74.77	± 11.05	76.84	± 9.73	86.40	± 9.03	74.60	± 8.88
Arylation	BO-ICL	5	74.77	± 9.89	83.65	± 2.42	87.93	± 5.82	76.53	± 5.33
Arylation	LA-MCTS	5	74.77	± 11.05	78.37	± 7.12	80.51	± 9.37	73.70	± 8.19
Arylation	Atlas only	5	74.77	± 11.05	78.98	± 7.75	90.44	± 6.70	76.18	± 9.13
Arylation	BotorchQEI only	5	74.77	± 11.05	81.11	± 8.74	88.91	± 7.40	77.20	± 7.67
Arylation	TRACE + Atlas	5	69.47	± 7.70	85.33	± 14.62	94.37	± 5.00	77.11	± 6.90
Arylation	TRACE + BotorchQEI	5	73.41	± 12.52	87.36	± 7.93	<u>93.67</u>	± 4.13	81.57	± 5.85
Suzuki HTE	Random	5	70.39	± 24.83	90.32	± 5.97	93.06	± 6.08	83.78	± 9.40
Suzuki HTE	BO-ICL	5	70.39	± 22.21	88.93	± 5.10	96.47	± 2.63	84.42	± 8.46
Suzuki HTE	LA-MCTS	5	70.39	± 24.83	90.71	± 8.66	91.09	± 8.88	80.71	± 16.25
Suzuki HTE	Atlas only	5	70.39	± 24.83	75.86	± 29.15	93.31	± 7.05	78.85	± 21.72
Suzuki HTE	BotorchQEI only	5	70.39	± 24.83	86.05	± 17.15	91.93	± 11.67	82.79	± 17.32
Suzuki HTE	TRACE + Atlas	5	77.60	± 21.29	92.83	± 8.00	<u>96.37</u>	± 2.99	<u>86.41</u>	± 9.79
Suzuki HTE	TRACE + BotorchQEI	5	81.80	± 13.71	<u>90.88</u>	± 4.81	93.88	± 2.68	87.26	± 5.97
Buchwald sub1	Random	5	83.03	± 12.33	83.03	± 12.33	89.13	± 5.33	83.46	± 9.64
Buchwald sub1	BO-ICL	5	83.03	± 11.03	88.78	± 5.63	93.46	± 4.46	86.02	± 7.16
Buchwald sub1	LA-MCTS	5	83.03	± 12.33	90.62	± 8.38	96.12	± 2.54	88.38	± 7.30
Buchwald sub1	Atlas only	5	83.03	± 12.33	88.85	± 9.89	93.93	± 5.69	86.68	± 9.05
Buchwald sub1	BotorchQEI only	5	83.03	± 12.33	87.49	± 14.48	95.63	± 5.75	<u>86.92</u>	± 9.75
Buchwald sub1	TRACE + Atlas	5	83.02	± 10.41	90.85	± 8.38	97.23	± 2.53	84.89	± 6.08
Buchwald sub1	TRACE + BotorchQEI	5	74.07	± 16.03	95.96	± 2.82	<u>96.78</u>	± 2.67	86.26	± 3.29
Buchwald sub2	Random	5	48.67	± 7.93	55.39	± 6.48	55.39	± 6.48	49.16	± 5.71
Buchwald sub2	BO-ICL	5	48.67	± 7.09	52.11	± 6.48	55.77	± 4.53	48.51	± 6.51
Buchwald sub2	LA-MCTS	5	48.67	± 7.93	49.97	± 7.33	58.25	± 5.83	49.53	± 4.73
Buchwald sub2	Atlas only	5	48.67	± 7.93	51.68	± 6.00	55.14	± 6.02	48.48	± 6.33
Buchwald sub2	BotorchQEI only	5	48.67	± 7.93	51.72	± 10.14	58.03	± 10.08	49.26	± 10.13
Buchwald sub2	TRACE + Atlas	5	49.00	± 12.54	57.31	± 9.48	<u>59.22</u>	± 8.11	<u>53.45</u>	± 5.30
Buchwald sub2	TRACE + BotorchQEI	5	45.89	± 7.10	<u>55.56</u>	± 8.11	63.03	± 7.00	54.09	± 4.26

Table A2: Full quantitative results over five seeds. Means are shown with gray standard deviations. **Bold** and underline mark the best and second-best values within each dataset for best@10, best@20, and AUC; best@5 is reported as an initialization diagnostic.

Field	Recorded value
Iteration / observation	iter 8 / obs 13
Task / optimizer	Buchwald sub2 / BoTorch qEI
Optimizer proposal	qEI top- k shortlist
Diagnosis	uneven coverage; Additive underexplored
Selected action	shape_only_bo_pick
Action rationale	rebalance coverage while staying within the qEI shortlist
Skill invoked	shortlist coverage shaping
Evidence access	reviewed evidence used (count 4)
Candidate source	bo_top1_from_shaped_shortlist
Candidate probe	true; differs from optimizer top-1 candidate
Outcome	result 66.016; best-so-far 57.089 $\rightarrow$ 66.016

Table A4: Example trace artifact from a Buchwald sub2 qEI run (seed 8, iteration 8). Each row maps to a component of the per-iteration record τ_t .

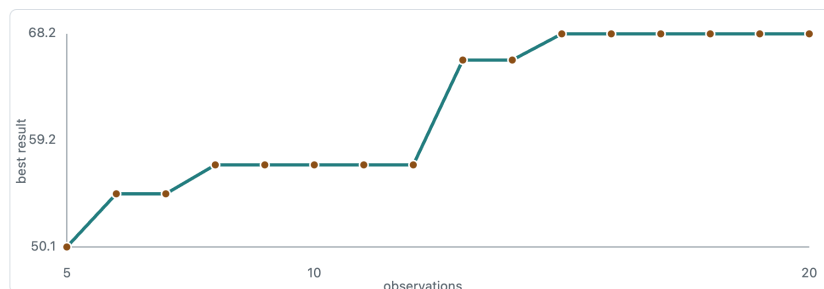
Iter.	Selected action	Evidence	Candidate probe	Differs from optimizer top-1	Result	Best-so-far
5	shape_only_bo_pick	yes (4)	true	true	16.721	57.089
6	shape_only_bo_pick	yes (2)	false	false	53.384	57.089
7	shape_only_bo_pick	yes (4)	false	false	53.335	57.089
8	shape_only_bo_pick	yes (4)	true	true	66.016	66.016

Table A5: Compact audit example from the Buchwald sub2 qEI run shown in Figures A4 and A5.

Run Summary

Dataset buchwald_sub2	Planner botorch_qei	Seed 8	Budget 20 total / 5 init	Best@5 50.104	Best@10 57.089	Best@20 68.248
Final best 68.248	LLM calls 82	LLM tokens 1280399	LLM cost \$1.0393	LLM model gpt-5.4-mini		

Best-so-far Trajectory



Action Timeline

Iter 1 direct_bo_pick Improved: True result 54.632 best 54.632 knowledge: no (0) candidate probe: False changed BO top1: False No controller intervention triggered.
Iter 2 direct_bo_pick Improved: False result 51.397 best 54.632 knowledge: no (0) candidate probe: False changed BO top1: False No controller intervention triggered.

Figure A4: Static audit view overview for a Buchwald sub2 run with the BoTorch qEI optimizer. The view summarizes the run, plots the best-so-far trajectory, and lists the action timeline generated from the stored run artifacts.

Item type	Scope	Intended stage	Operational role	Safety note
Reviewed domain knowledge	Suzuki HTE	initialization	span distinct activation regimes rather than near-neighbor conditions	abstract design guidance, not candidate ranking
Reviewed domain knowledge	Buchwald sub1	initialization	balance high-quality anchors with controlled coverage	avoids exact answer tuples or exact additive rules
Reviewed experience	Arylation	initialization diagnosis	maintain ligand diversity when base coverage dominates the slate	used as an axis-balance guardrail
Procedural skill	general	runtime diagnosis	probe candidate coverage and optimizer shortlist quality	called only under scoped actions
Procedural skill	general	offline governance	build and review evidence cards from documents or traces	reviewed before runtime activation

Table A6: Abbreviated Evidence & Skill System examples. Evidence provides scoped support for a decision; skills define executable or governance procedures available to the agent.

Decision Detail

► Iteration 1: direct_bo_pick -> result 54.632, best 54.632

► Iteration 2: direct_bo_pick -> result 51.397, best 54.632

► Iteration 3: direct_bo_pick -> result 57.089, best 57.089

► Iteration 4: direct_bo_pick -> result 47.953, best 57.089

► Iteration 5: shape_only_bo_pick -> result 16.721, best 57.089

► Iteration 6: shape_only_bo_pick -> result 53.384, best 57.089

► Iteration 7: shape_only_bo_pick -> result 53.335, best 57.089

▼ Iteration 8: shape_only_bo_pick -> result 66.016, best 66.016

Diagnosis summary none Coverage: strong key-dimension coverage with a single weak dimension; not stagnating	Action reasoning / plan The run is not truly stagnating: the incumbent recently improved to 57.09, duplicate ratio is 0.0, and only 12/792 candidates have been visited. However, coverage is highly uneven, with Reactant2, Ligand, and Base saturated in the visited set while Additive remains the only clearly underexplored dimension. That means the correct intervention is not controll...	Selected candidate Reactant2 <chem>C0c1ccc(I)cc1</chem> Ligand <chem>C0c1ccc(OC)c(P([C@]23C[C@H]4C[C@H](C[C@H](C4)C2)C3)[C@]23C[C@H]4C[C@H](C[C@H](C4)C2)C3)c1-c1c(C(C)C)cc(C(C)C)cc1C(C)C</chem> Base <chem>CCN=P(N=P(N(C)C)(N(C)C)N(C)C)(N(C)C)N(C)C</chem> Additive <chem>c1ccc(CN(Cc2ccccc2)c2ccon2)cc1</chem>
BO top1 candidate Reactant2 <chem>C0c1ccc(I)cc1</chem> Ligand <chem>C0c1ccc(OC)c(P([C@]23C[C@H]4C[C@H](C[C@H](C4)C2)C3)[C@]23C[C@H]4C[C@H](C[C@H](C4)C2)C3)c1-c1c(C(C)C)cc(C(C)C)cc1C(C)C</chem> Base <chem>CCN=P(N=P(N(C)C)(N(C)C)N(C)C)(N(C)C)N(C)C</chem> Additive <chem>CCOC(=O)c1cc(OC)no1</chem>	Knowledge used=True, count=4 N/A	Guardrail / fallback reason override_guardrail_disabled
Reflection / what we learned The latest candidate materially improved the incumbent to 66.02, so the search is still active rather than stalled; however, the improvement came from the same anchored Buchwald-Hartwig neighborhood and does not invalidate the earlier coverage diagnosis. Reactant2, Ligand, and Base remain sufficiently explored relative to Additive, which still looks like the main information bottleneck and likely mediates the observ...		

Figure A5: Expanded audit record for iteration 8. The view exposes the diagnosis, action reasoning, selected candidate, optimizer top candidate, evidence use, guardrail or fallback status, and reflection associated with the observed improvement.